

Quantum Field Theory – Homework 7

Name: _____ Due Date: 2025.12.08

These problems are based on Lectures 13 and 14, focusing on the derivation of path integrals from time-slicing, the short-time kernel, the BCH approximation, and the use of the path integral to compute time-ordered correlation functions. Show key steps clearly and state any assumptions you use.

Problem 1. Short-Time Kernel and the Lagrangian (15 points)

In Lecture 13, the short-time propagator was obtained by approximating

$$e^{-i\hat{H}\Delta t} \simeq e^{-i\Delta t \frac{\hat{p}^2}{2m}} e^{-i\Delta t V(\hat{x})} [1 + \mathcal{O}(\Delta t^2)],$$

using the Baker–Campbell–Hausdorff (BCH) formula.

- (a) Starting from the operator identity

$$e^{A+B} = e^A e^B \exp\left(-\frac{1}{2}[A, B] + \dots\right),$$

show explicitly that the corrections from the commutator $[\hat{p}^2, V(\hat{x})]$ are of order $\mathcal{O}(\Delta t^2)$ when deriving the short-time kernel. State clearly the scaling assumption for Δt .

- (b) Evaluate the matrix element

$$\langle x_{i+1} | e^{-i\Delta t \frac{\hat{p}^2}{2m}} | x_i \rangle$$

by inserting a complete set of momentum eigenstates. Show that the result is

$$\sqrt{\frac{m}{2\pi i \Delta t}} \exp\left[\frac{im}{2} \left(\frac{x_{i+1} - x_i}{\Delta t}\right)^2 \Delta t\right].$$

- (c) Combine this with $\langle x_{i+1} | e^{-i\Delta t V(\hat{x})} | x_i \rangle = e^{-i\Delta t V(x_i)} \delta(x_{i+1} - x_i)$ to show that the full short-time propagator is

$$\langle x_{i+1} | e^{-i\hat{H}\Delta t} | x_i \rangle = \sqrt{\frac{m}{2\pi i \Delta t}} \exp\left[i\Delta t \left(\frac{1}{2}m\dot{x}_i^2 - V(x_i)\right)\right],$$

where $\dot{x}_i \equiv (x_{i+1} - x_i)/\Delta t$.

Problem 2. Path Integral and Vacuum Projection (15 points)

In Lecture 14 we used the “ $i\epsilon$ trick” to project onto the ground state in the limit $t \rightarrow +\infty$, $t' \rightarrow -\infty$.

- (a) Let $\{|n\rangle\}$ be the energy eigenbasis of a Hamiltonian H . Show that inserting a factor

$$e^{-iH(1-i\epsilon)t}$$

suppresses the contribution of all excited states as $t \rightarrow +\infty$, and derive explicitly that

$$\lim_{t \rightarrow +\infty} \langle x, t | n \rangle \propto e^{-E_n \epsilon t}.$$

Explain why only the ground state survives.

- (b) Using the result of part (a), show that for an operator X

$$\lim_{t \rightarrow \infty, t' \rightarrow -\infty} \langle x, t | X | x', t' \rangle = \psi_0(x) \psi_0^*(x') e^{-iE_0(t-t')} \langle 0 | X | 0 \rangle,$$

where $\psi_0(x) = \langle x | 0 \rangle$.

- (c) Explain why taking the ratio

$$\frac{\langle x, t | X | x', t' \rangle}{\langle x, t | x', t' \rangle} \quad \text{as } t \rightarrow +\infty, t' \rightarrow -\infty$$

removes the wavefunctions and the divergent normalization constants, yielding

$$\langle 0 | X | 0 \rangle.$$

- (d) Apply this to the operator insertion $X = T(x(t_1) x(t_2))$ and write the path-integral expression for the two-point function

$$G_2(t_1, t_2) = \langle 0 | T(x(t_1) x(t_2)) | 0 \rangle.$$

Problem 3. Bonus (optional): From QM to QFT (5 points)

Explain why the replacement

$$x(t) \longrightarrow \phi(t, \mathbf{x}), \quad \int dt \longrightarrow \int d^4x,$$

together with the identification

$$S[\phi] = \int d^4x \left(-\frac{1}{2} \partial_\mu \phi \partial^\mu \phi - V(\phi) \right),$$

leads directly from the QM path integral to the QFT functional integral

$$Z = \int \mathcal{D}\phi e^{iS[\phi]}.$$

Which conceptual step from Lecture 14 guarantees that fields at different spatial points act like infinitely many decoupled degrees of freedom in the free theory?